DarkTies: Detecting APT-Affiliated Actors in Underground Forums

Design Proposal

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1 Refined Statement of the Research Problem

1.1 Introduction to the Research Problem

Underground hacking forums are a documented source of early warning signals for cyber threats: analysis of 88 million forum posts has shown that more than 60% of threat-relevant keywords appear in forum discussions before they appear in professional security reports [6]. Advanced Persistent Threat (APT) groups, however, do not use these forums as ordinary participants. Industry investigations have identified forum usernames of nearly twenty known APT groups, whose forum activity is narrowly purposive: recruiting partners, selling malware-as-a-service and network access, and trading compromised data [1]. APT behavioral research in peer-reviewed papers profiles these groups from historical incidents and tactics rather than from forum activity [12], and APT attribution research relies on technical evidence such as malware samples [21]. The intersection — behavioral analysis of forum users for the purpose of detecting affiliation with known APT actors — is therefore largely unoccupied in the academic literature. A system operating in this space would answer a question that neither forum-mining research nor APT-profiling research currently answers: *which members of the wider underground population are likely cooperating with, or belonging to, known APT groups?*

This project proposes the design, implementation, and evaluation of an **agentic system** that ingests underground forum data, constructs interaction graphs over forum personas, infers communities and user roles using methods established in the literature, and produces a graded *APT-affiliation score* for each persona, seeded by known APT-linked usernames. The system is passive: it analyzes already-collected forum corpora and does not interact with forum users.

The central methodological difficulty is that *affiliation* is not a single claim, and that *ground truth* is scarce and partly probabilistic. We therefore decompose the target construct into three levels, each with a different evidence threshold:

1. **Identity match (L1)** — two nicknames refer to the same person; testable with username-linkage and stylometric methods [20].
2. **Cooperation (L2)** — two personas interact in a way consistent with operational collaboration; a behavioral claim tested on interaction and content signals.
3. **Membership (L3)** — a persona belongs to a named APT group; the strongest and scarcest claim, testable only against gold-tier evidence (Section 1.2).

1.2 Ground-Truth Evidence Tiers

The three levels above state *what kind* of affiliation is claimed. A separate question is *how well* a given claim can be evidenced, because hard ground truth for APT affiliation is scarce. The project therefore grades every label into one of three confidence tiers, each with documented provenance:

- **Gold**: law-enforcement-endorsed links between forum handles and real actors (indictments, prosecution records) and, for the membership level, internal group records such as the Conti leaks (February 2022), which map handles to organizational roles.
- **Silver**: vendor-attributed personas, e.g., the APT forum usernames identified by industry investigation [1].
- **Bronze**: entity-level evidence (mentions of APT tooling and actor names), used only to anchor the entity-recognition layer, not as persona labels.

Every hypothesis and experiment in this proposal states which tiers it relies on, and every reported result is traceable to its evidence tier.

1.3 Research Questions

From this decomposition we derive four research questions:

RQ1 Can the behavioral and structural footprint of a forum persona distinguish APT-affiliated actors from the general underground population?

RQ2 Which signals — community position, inferred role, interaction proximity to seed personas, or textual signals extracted by language models from the users' posts — carry the affiliation signal, and how do they combine?

RQ3 Does a multi-agent LLM pipeline — in which agents adaptively decide, for each persona, which signals to inspect and how to weigh them — rank APT-affiliated personas more accurately than fixed, non-agentic scoring methods built from the same signals?

RQ4 Can the system surface affiliated personas *before* their public attribution, providing actionable early warning?

Scope delimitation. The project operates on archived research corpora (Section 4); live scraping is an optional, ethics-approved extension, not the core method. All outputs are probabilistic risk scores over personas, not assertions over persons, and no named individuals are published.

2 Current State of the Literature Analysis

The literature analysis supporting this proposal is organized in three threads.

2.1 Reviews and systematizations

The journal-level literature is young, fragmented, and concentrated in a small set of publication outlets. Nehme et al. systematize the processing of hacker forum data into cyber threat intelligence (CTI) artifacts using an input–process–output model [22]; Cascavilla et al. map cybercrime threat intelligence across the surface, deep, and dark web [3]; Ignaczak et al. survey text mining in cybersecurity [4]; and Basheer and Alkhatib review dark web investigation research for CTI [5]. Two conclusions from these reviews directly motivate this proposal: (i) there is no standard benchmark or evaluation protocol for behavioral forum analysis, and (ii) ethical and legal constraints are repeatedly acknowledged but not standardized. Both are gaps the project can partially close by publishing its evaluation protocol.

2.2 Building blocks for the proposed system

Four method families provide validated components. First, *community detection*: Manatova et al. systematically evaluate community-detection methods on hacker forums and find that neighborhood-based methods (e.g., Expansion) are best suited to CTI, and propose an investigation workflow [11]. Second, *role and expertise inference*: Biswas et al. classify forum users into expertise tiers and show that expert hackers behave as leaders of communities of practice [9], and HackerRank combines content analysis with topic-specific PageRank to identify key hackers across five underground forums [8]. Third, *actor-level CTI platforms*: the AZsecure Hacker Assets Portal demonstrates a deployed academic system that collects, analyzes, and reports on forum and marketplace data [10], and Pastrana et al. provide the large-scale behavioral characterization of forum actors that defines the genre [18], a lineage continued by recent work that identifies key expert actors in cybercrime forums by their technical expertise [19]. Fourth, *entity tracking and event detection*: Paladini et al. track threat-actor and malware names across 88 million posts with domain-adapted BERT models [6], and EventHunter dynamically clusters and

ranks security events from forum discussions [7]. Together these components cover the ingestion, graph-construction, community-detection, and role-inference stages of the proposed pipeline.

2.3 Machine-learning and agentic elements

For the language-model layer, DarkBERT provides a domain-adapted pretrained model for dark web text [13], and Prado-Sánchez et al. benchmark eight commercial LLMs in zero-shot classification of illicit dark web content, documenting accuracy, human consistency, and inter-model agreement [14]. On the agentic side, MAD-CTI is a journal-published, AutoGen-based multi-agent framework that automates scraping, analysis, and classification of dark web content, with open-source code [15]; Clairoux-Trepanier et al. demonstrate zero-shot extraction of structured CTI variables from conversations on the XSS, Exploit.in, and RAMP forums with high precision on initial-access detection [17]; and DarkBot extends the frontier to active elicitation, with correspondingly sharper ethical questions [16]. These works establish that agentic forum processing exists; none of them performs APT-affiliation analysis, which is where this project positions its contribution. The novelty therefore rests on the affiliation layer and its evaluation protocol, not on the agent architecture itself.

2.4 The APT gap

Industry research confirms that APT groups maintain forum personas [1], yet the peer-reviewed APT-profiling literature builds group profiles from incidents and tactics rather than forum behavior [12], and attribution research works from technical artifacts [21]. No peer-reviewed work isolates named APT groups as a behavioral-analysis population on hacker forums, and no published system (agentic or otherwise) scores forum users for likely cooperation with known APT members. This is the gap the project fills.

3 Research Hypotheses

H1 (Discrimination). Affiliated personas can be separated from ordinary forum users. Personas with confirmed APT affiliation — the gold and silver tiers of Section 1.2 — receive higher affiliation scores than matched non-affiliated personas and appear at the top of the ranking more often than a random reshuffling of the labels would predict.

H2 (Lead time). The system flags affiliated personas before their affiliation becomes public knowledge. For personas whose affiliation became public only after the evaluation cutoff date, the system places them in its top-k flagged set before that public date; the success measure is the median lead time between the first flag and the public attribution, which should be strictly greater than zero.

H3 (Agentic advantage). Adaptive signal selection outperforms fixed formulas. The full multi-agent pipeline, in which agents decide for each persona which signals to inspect and how to weigh them, ranks affiliated personas more accurately than each non-agentic baseline built from the same signals — interaction-graph proximity to seed personas, content similarity over domain embeddings, and centrality — when compared at matched false-positive budgets.

H4 (Component necessity). Both signal families are needed. Removing either the community-detection layer or the role-inference layer significantly degrades ranking performance, which would show that structural signals (who interacts with whom) and behavioral signals (what role a persona plays) each carry information that the other cannot replace.

H5 (Transfer). The affiliation signal is not an artifact of a single community. A model calibrated on one source forum retains above-chance ranking performance on a held-out forum from the same corpus.

Table 1 maps the hypotheses to research questions and to the experiments of Section 4.

Table 1: Mapping of hypotheses to research questions and experiments.

Hypothesis	RQ	Experiment	Primary outcome
H1	RQ1, RQ2	E2b	Top-k precision, PR curve
H2	RQ4	E3	Median lead time
H3	RQ3	E4a	Ranking metrics vs. baselines
H4	RQ2	E4b	Ablation deltas
H5	RQ1	E5	Cross-forum ranking metrics

4 Plan of Experiments

The experiments are numbered in logical order of the evaluation argument; the only deliberate deviation in execution is E4a, which is run early as a design check, as noted in its subsection.

4.1 Data and ground-truth protocol (E1)

The evaluation runs on archived forum corpora — CrimeBB, made available by the Cambridge Cybercrime Centre under a data-sharing agreement, and/or the AZsecure hacker forum archives — because these collections span multiple years and allow looking back in time. Positive labels follow the three confidence tiers defined in Section 1.2, and the provenance of every label (indictment, leak record, or vendor report) is documented, so that each result states the strength of the evidence behind it. Because even silver labels can be wrong — the same handle may be reused by different people on different forums — every analysis is repeated under a worst-case assumption that a fixed fraction of the labels is incorrect. The evaluation protocol is written down and frozen before the final runs, so that no design choice can be tuned to the answers it produces.

4.2 Component-wise validation and affiliation ranking (E2)

E2 is split into two parts: the pipeline stages are first validated separately (E2a), and the full system is then evaluated on its main task, ranking personas for affiliation (E2b).

4.2.1 Component-wise validation (E2a)

Each pipeline stage is first tested on its own, against subtasks for which denser ground truth exists: role and expertise detection against the framework of Biswas et al. [9]; key-actor ranking against HackerRank [8]; related-account identification following the username-linkage methodology of Perito et al. [20]. Testing the stages separately shows where error enters the chain before it can reach the affiliation score.

4.2.2 Affiliation ranking (E2b, tests H1)

The system outputs a ranked list of personas, from the most to the least likely to be APT-affiliated; the question is whether the truly affiliated ones rise to the top. We measure top-k precision — the fraction of the first 10, 50, and 200 flagged personas that is subsequently corroborated — as well as precision–recall curves and nDCG against seed actors. Significance is

assessed with permutation tests, which repeatedly shuffle the labels and ask how often a random ranking would look this good. The system is deliberately evaluated as a ranking rather than a binary classifier, because affiliated personas are rare and a classifier would score mostly on the abundant negatives.

4.3 Temporal holdout and lead time (E3, tests H2)

The labels are split by time, not at random. The system is tuned only on personas attributed before a cutoff date and shown only data from before that date; it is then evaluated on personas whose affiliation became public after the cutoff. For each such persona, the lead time is measured: how long before the public attribution date the system first placed the persona in its top-k flagged set. This mirrors real deployment, where the system must flag actors the public does not yet know about, and it removes circularity, because the test labels did not exist when the system made its decisions.

4.4 Baselines and ablations (E4, tests H3 and H4)

E4 is split into two parts: the pipeline is compared with fixed, non-agentic scoring methods (E4a), and its components are removed one at a time (E4b).

4.4.1 Baselines (E4a, tests H3)

The full pipeline is compared with three fixed, non-agentic scoring methods built from the same raw signals: (i) proximity to seed personas in the interaction graph (shortest path, common neighbors), (ii) content similarity using domain embeddings [13], and (iii) centrality scores. This tests whether the adaptive decisions of the agents add anything over fixed formulas (H3). The comparison is run early for a deliberate reason: if a trivial proximity heuristic matches the full pipeline, the design must be revised.

4.4.2 Ablations (E4b, tests H4)

In separate runs, one pipeline component at a time is removed — first the community-detection layer, then the role-inference layer — and the ranking is recomputed. A significant performance drop after a removal shows that the removed component carries real information; this tests whether both signal families are necessary (H4).

4.5 Cross-forum transfer (E5, tests H5)

The model is calibrated on one source forum and evaluated, unchanged, on a different, held-out forum from the corpus. This tests whether the affiliation signals are general properties of underground communities or artifacts of one particular forum (H5). Some degradation is expected; the success criterion is that ranking performance on the new forum remains above chance.

4.6 Robustness analyses (E6)

Two robustness checks accompany the main results. First, *semi-synthetic injection*: real APT communication traces (adapted from leaked group chats) are injected into the corpus under altered surface features, and detection sensitivity to them is measured. These results are reported separately and never as the headline finding, because the system may detect the traces of synthesis rather than genuine affiliation. Second, *label-noise sensitivity*: every conclusion must survive a worst-case analysis in which a fixed fraction of the labels is assumed to be incorrect.

4.7 Analyst adjudication (E7)

The ranking metrics of E2b measure how high the known affiliates appear on the list, but they cannot judge the flags the system produces on personas for which no label exists. E7 answers that complementary question with human judgment. A random sample of personas flagged in the main runs is drawn, and for each persona an evidence dossier is compiled: vendor reporting, later indictments, and overlap of tooling or infrastructure with known APT activity. At least two independent annotators then review each dossier, blind to the system's score, and each assigns one of three verdicts: affiliated, inconclusive, or not affiliated. Their agreement is reported with Cohen's κ, which corrects for agreement by chance ($\kappa = 1$ means perfect agreement; $\kappa = 0$ means agreement no better than chance); disagreements are resolved by discussion or, if they persist, by a third annotator. The fraction of adjudicated personas judged affiliated is a human-validated estimate of the system's precision, and the cases the annotators reject provide a qualitative error analysis of what the system mistakes for affiliation.

4.8 Ethical and legal framework

Darknet research that studies or is likely to discover illegal activity requires ethics-committee approval, and the ethics application precedes implementation [2]. The core experiments use archived research corpora only; the system is passive and does not interact with forum users; and the paper publishes no named individuals — outputs are reported as aggregate metrics and anonymized examples. The proposal includes an explicit risk-of-harm analysis: mislabeling a pseudonymous user as APT-affiliated can expose that person to retaliation or state action, which is why all outputs are calibrated risk scores with uncertainty, never definitive claims.

4.9 Threats to validity

- **Label noise and pseudonymity**: a handle is not a person; mitigated by the three-level construct, tiered labels, and the worst-case analysis of E6.
- **Corpus recency**: the archived forums predate the migration of the underground economy to Telegram and Discord; claims are scoped to archival data, and the limitation is stated explicitly.
- **Circularity**: the risk that the system is tuned, knowingly or not, on the very personas it is later evaluated on, so that the results look good by construction. Mitigated by pre-registering the protocol and freezing it before the final runs, and by the temporal holdout of E3, in which the test labels did not exist when the system made its decisions.
- **Synthesis artifacts**: mitigated by separating the injection study (E6) from the headline evaluation.
- **Statistical power**: the positive class is small; the design therefore relies on ranking metrics, permutation tests, and accumulated evidence across E2–E7 rather than a single train/test split.

4.10 Expected contributions

The project is expected to deliver the following contributions:

(1) the first peer-reviewed system for APT-affiliation analysis of underground forum users;

(2) a documented, reusable evaluation protocol for LLM-driven behavioral analysis of underground forums, partially filling the benchmark gap identified in the reviews;

(3) empirical evidence on the relative contribution of structural, behavioral, and content signals to the affiliation task.

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